EDS 223: Homework 2

Mapping HOLC Neighborhoods in Los Angeles

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Background

In 1933 the Home Owners' Loan Act was adopted, and led to the establishment of the Home Owners' Loan Corporation (HOLC) to help homeowners that were facing foreclosure by refinance mortgages [1]. The HOLC created a mortgage lending risk classification system, and appraisers were tasked with surveying and mapping neighborhoods. Appraisers evaluated housing age, condition, transportation access, proximity to hazards or amenities, and economic, racial, and ethnic composition of residents. Neighborhoods were assigned to a grade based on loan approval level: green for "Best", blue for "Still Desirable", yellow for "Definitely Declining", and red for "Hazardous". Green and blue grades were typically assigned to predominantly white neighborhoods, and had high home loan approvals. Neighborhoods that were predominately composed of Black Americans and immigrants were redlined, and denied home loans [2]. These practices limited access to homeownership and contributed to long-term racial, economic, and environmental inequalities.

A recent study examined the correlation between structural racism in historically redlined neighborhoods in Boston, MA and pedestrian fatalities between 2010 to 2019. Results showed that pedestrian fatalities were highest in historically redlined neighborhoods, as HOLC grade decreased there was an increase in pedestrian fatality rate [3].

This analysis examines the relationship between historical HOLC neighborhood grades and recent socioeconomic conditions, environmental indicators, and bicycle and pedestrian crashes and fatalities in Los Angeles, California.

Data

Historical HOLC data was originally downloaded from University of Richmonds' Mapping Inequality project.

Bike and pedestrian accident data came from the Statewide Integrated Traffic Records System (SWITRS) database. Census block group data from the Environmental Protection Agency's Environmental Justice Screening and Mapping Tool (EJSCREEN). This data set combines environmental and demographic indicators.

```r
# Reading in data
ejscreen <- sf::st_read(here::here(
  "data",
  "ejscreen",
  "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"
))
holc <- sf::st_read(here::here(
  "data",
  "mapping-inequality",
  "mapping-inequality-los-angeles.json"
))
crashes <- read_csv(here::here(
  "data",
  "SWITRS_LA",
  "Crashes.csv"
))
```

Methods

Historical HOLC neighborhood data was transformed to match the Coordinate Reference System (CRS) of the EJSCREEN data set. The EJSCREEN data set was filtered to Los Angeles County, and st_centroid() was used to convert each census block group polygon to a single point representing its geometric center. The HOLC data was spatially joined with the census data to assign each census block group the grade of the HOLC polygon containing its centroid.

The SWITRS crash dataset, which contains crashes from 2020 to 2025, was examined for missing coordinate values to determine which variables to use for spatial analysis. The POINT_X and POINT_Y fields were used because they had far fewer missing values than the LATITUDE and LONGITUDE fields (9,484 vs. 97,350). Crashes with missing coordinates were removed, and the dataset was converted to a simple feature and transformed to match the EJSCREEN CRS. The crash dataset was filtered to bicycle and pedestrian crashes (including fatal crashes) and spatially joined with the historical HOLC polygons. Crashes without an HOLC grade were removed.

Historical HOLC grades in the city of Los Angeles were mapped and color coded using the HOLC classifications: green for “Best,” blue for “Still Desirable,” yellow for “Definitely Declining,” and red for “Hazardous.” The percentage of Los Angeles census block groups within each HOLC grade was calculated, including census block groups with no HOLC grade.

To examine low-income percentage by HOLC grade, census block groups without HOLC grades were removed from the dataset, and the mean low-income population percentage (LOWINCPCT) was calculated for each grade. The results were plotted on a bar graph and color coded by HOLC grade. To identify potential environmental justice concerns, the low life expectancy percentile (P_LIFEEXPPCT) and particulate matter 2.5 percentile (P_PM25) were selected from the cleaned census block group dataset, and their averages were calculated for each grade. These values were plotted on a scatter plot to visualize their relationship.

To identify differences in bicycle and pedestrian crashes in historical HOLC grade neighborhoods, the cleaned crash dataset was used to calculate the percentage of bicycle and pedestrian crashes by HOLC grade, which was plotted on a bar graph.

```
# Transforming the CRS
holc <- st_transform(holc, crs = st_crs(ejscreen))

# Verifying CRS match
if (st_crs(holc) != st_crs(ejscreen)) {
  stop("coordinate reference systems do not match")
} else {
  print("it's a match!")
}
```

```
[1] "it's a match!"
```

```
# Filtering to LA County
# Using st_centroid to assign one HOLC grade per census block group
la_cnty <- ejscreen |>
  filter(CNTY_NAME == "Los Angeles County") |>
  st_centroid()

# Joining census block groups and HOLC grade data
holc_census <- st_join(la_cnty, holc, join = st_within)

# Verifying join rows match
if (nrow(holc_census) == nrow(la_cnty)) {
  print("join matches original data dimensions")
} else {
  warning("join does not match original data dimensions")
  print(paste("holc_census has", nrow(holc_census), "rows"))
  print(paste("la_cnty has", nrow(la_cnty), "rows"))
}
```

```
[1] "join matches original data dimensions"
```

```
# Filtering out NA HOLC grade rows
holc_bg <- holc_census |>
  filter(!is.na(grade))
```

```r
# Checking missing coordinate values
sum(is.na(crashes$LATITUDE))
```

```
[1] 97350
```

```r
sum(is.na(crashes$LONGITUDE))
```

```
[1] 97350
```

```r
sum(is.na(crashes$POINT_X))
```

```
[1] 9484
```

```r
sum(is.na(crashes$POINT_Y))
```

```
[1] 9484
```

```r
# Filtering and transforming to simple feature and setting CRS to match ejscreen
crashes_sf <- crashes |>
  filter(!is.na(POINT_X) & !is.na(POINT_Y)) |>
  rename(long = POINT_X, lat = POINT_Y) |>
  st_as_sf(
    coords = c("long", "lat"),
    crs = 4326
  ) |>
  st_transform(
    crs = st_crs(ejscreen)
  )

# Verifying CRS match
if (st_crs(crashes_sf) != st_crs(ejscreen)) {
  stop("coordinate reference systems do not match")
} else {
  print("it's a match!")
}
```

```
[1] "it's a match!"
```

```r
# Checking years in data set
unique(crashes_sf$ACCIDENT_YEAR)
```

```
[1] 2020 2023 2021 2025 2024 2022
```

```r
# Filtering to bicycle and pedestrian accidents
bp_acc <- crashes_sf |>
  filter(
    BICYCLE_ACCIDENT == "Y" | PEDESTRIAN_ACCIDENT == "Y"
  )

# Joining accident data to HOLC grades
bp_holc <- st_join(bp_acc, y = holc, join = st_within)

# Verifying join row counts match
if (nrow(bp_holc) == nrow(bp_acc)) {
  print("join matches original data dimensions")
} else {
  warning("join does not match original data dimensions")
  print(paste("bp_acc_holc has", nrow(bp_holc), "rows"))
  print(paste("bp_acc has", nrow(bp_acc), "rows"))
}
```

```
[1] "join matches original data dimensions"
```

```r
# Removing NA HOLC grade
bp_holc <- bp_holc |>
  filter(!is.na(grade))
```

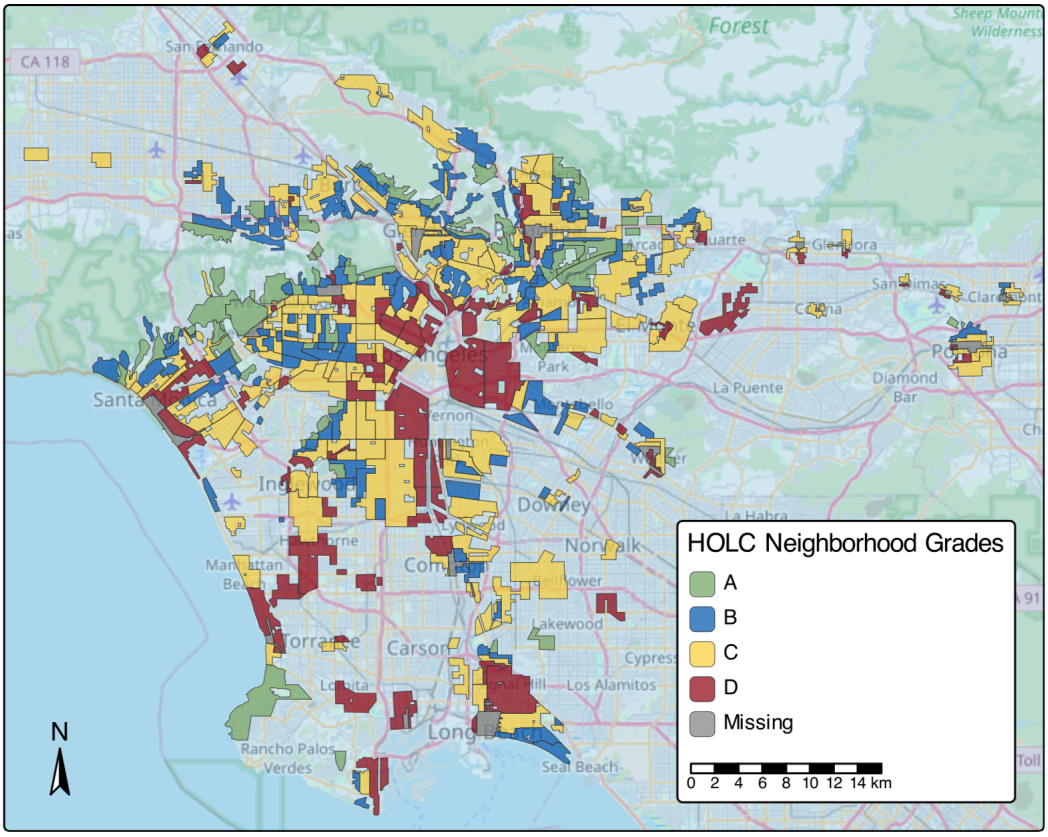
Results

```r
# Mapping HOLC grades in Los Angeles

hist_red <-
  tm_basemap("OpenStreetMap", zoom = 10, alpha = 0.55) +
  tm_shape(holc) +
  tm_polygons(
    fill = "grade",
    fill_alpha = 0.7,
    fill.scale = tm_scale_categorical(
```

```
      values = holc_colors,
      value.na = "grey50"
    ),
    col = "grey20",
    lwd = 0.3,
    fill.legend = tm_legend(
      title = "HOLC Neighborhood Grades",
      position = tm_pos_in("right", "bottom"),
      bg.color = "white"
    )
  ) +
  tm_compass(position = c("left", "bottom"), size = 1.5) +
  tm_scalebar(position = c("right", "bottom")) +
  tm_title("HOLC Neighborhood Grades in Los Angeles, CA (1939)") +
  tm_layout(
    text.fontfamily = "sans",
    style = "natural",
    outer.bg = TRUE,
    outer.bg.color = "white",
    frame = TRUE,
    legend.text.size = 0.65,
    legend.title.size = 0.8
  )
hist_red
```

HOLC Neighborhood Grades in Los Angeles, CA (1939)



The historical HOLC grades map shows a high proportion of lower-rated neighborhoods, Grades C (yellow) and D (red), in South LA, Boyle Heights and Long Beach. Higher-rated neighborhoods, Grades A (green) and B (blue), are seen in Pacific Palisades, Beverly Hills, West Los Angeles, and Palos Verdes.

```
# Grouping census block groups by HOLC grade
# Calculating percentage of census block groups per HOLC grade
# Creating table of summary
holc_cens_summ <- holc_census |>
  st_drop_geometry() |>
  group_by(grade) |>
  summarise(n_bg = n()) |>
  mutate(
    pcnt_cens_holc = (n_bg / sum(n_bg)) * 100,
    grade = ifelse(is.na(grade), "No HOLC grade", grade)
  ) |>
  select("grade", "pcnt_cens_holc") |>
```

```
  gt() |>
  tab_header(title = md("Census Block Groups Per HOLC Grade")) |>
  cols_label(
    grade = "HOLC Grade",
    pcnt_cens_holc = "Percent of Census Block Groups"
  ) |>
  fmt_number(
    columns = pcnt_cens_holc,
    decimals = 2,
    pattern = "{x}%"
  )
holc_cens_summ
```

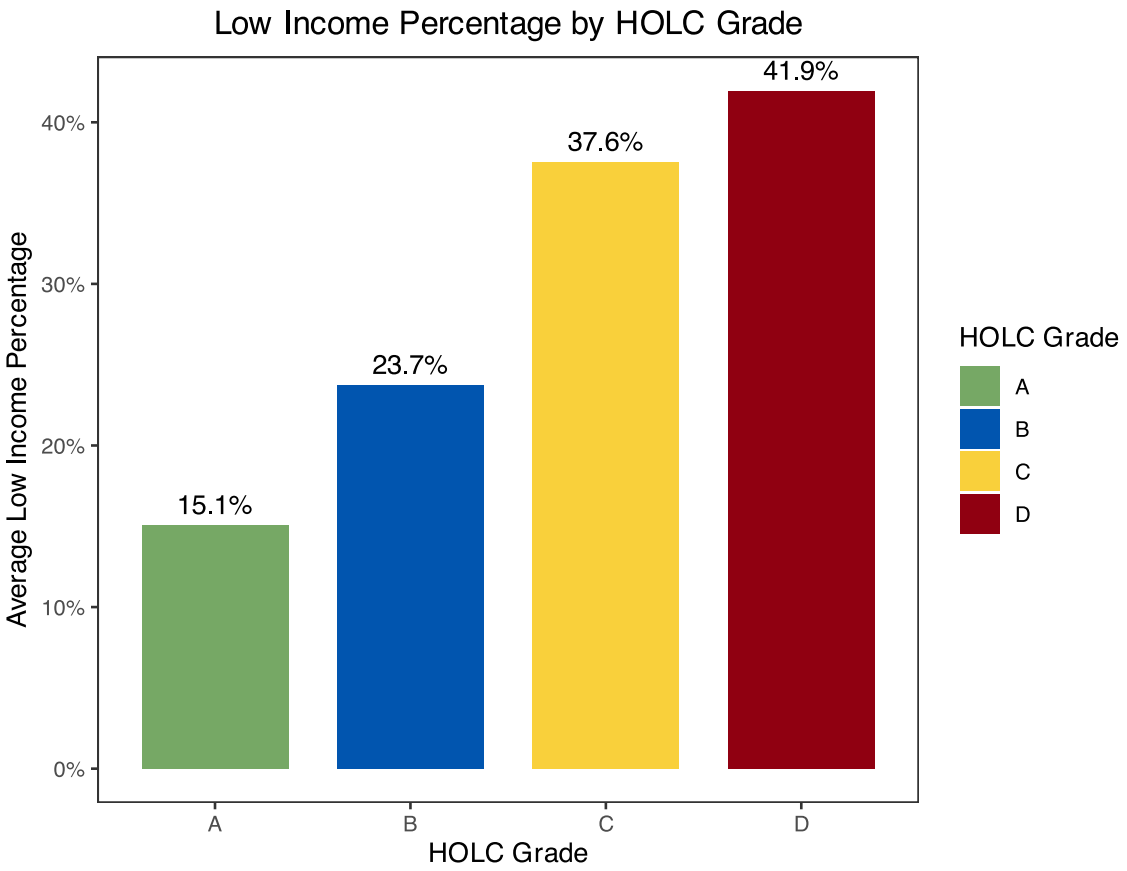
Census Block Groups Per HOLC Grade

HOLC Grade	Percent of Census Block Groups
A	2.43%
B	7.65%
C	23.68%
D	10.98%
No HOLC grade	55.26%

Of the census block groups with an associated HOLC grade, grade C had the highest percentage at 23.68%, followed by grade D (10.98%), grade B (7.65%), and grade A (2.43%). Census block groups that were not assigned an HOLC grade accounted for 55.26% of census block groups.

```
# Creating bar graph of mean low income percentage by HOLC grade
hc_low_income <- holc_bg |>
  st_drop_geometry() |>
  group_by(grade) |>
  summarise(mean_lowinc = mean(LOWINCPCT, na.rm = TRUE)) |>
  mutate(mean_lowinc_pct = mean_lowinc * 100) |>
  ggplot(
    aes(
      x = grade,
      y = mean_lowinc_pct,
      fill = grade
    )
  ) +
  geom_col(width = 0.75) +
```

```
  geom_text(
    aes(label = paste0(round(mean_lowinc_pct, 1), "%")),
    vjust = -0.5
  ) +
  scale_fill_manual(
    values = holc_colors
  ) +
  scale_y_continuous(labels = scales::label_percent(scale = 1)) +
  labs(
    title = "Low Income Percentage by HOLC Grade",
    x = "HOLC Grade",
    y = "Average Low Income Percentage",
    fill = "HOLC Grade"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5),
    panel.grid = element_blank()
  )
hc_low_income
```



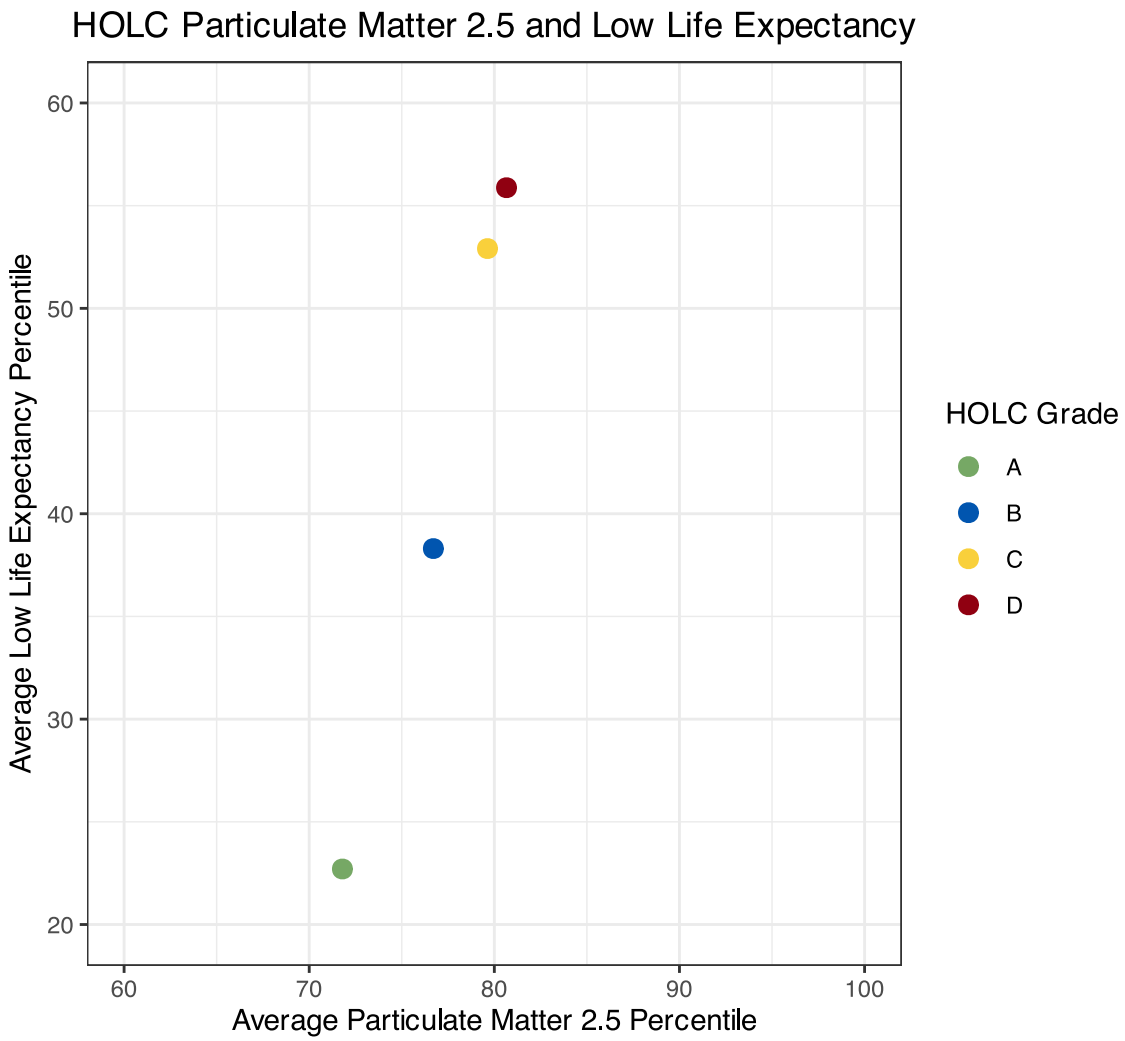
Average low-income population percentages increased as historical HOLC grades decreased. Neighborhoods in historical HOLC A and B grades had a lower average low income percentage compared to the neighborhoods in the historical C and D grades. Grade D had the highest low-income percentage at 41.9% and Grade A had the lowest percentage at 15.1%.

```
# Summarizing mean particulate matter 2.5 percentile by HOLC grade
pm25_lle <- holc_bg |>
  st_drop_geometry() |>
  group_by(grade) |>
  summarise(
    mean_pm25 = mean(P_PM25, na.rm = TRUE),
    mean_lle = mean(P_LIFEEXPPCT, na.rm = TRUE)
  )

# Scatter plot of mean particulate matter and low life expectancy percentiles
by HOLC grade

ggplot(
  data = pm25_lle,
```

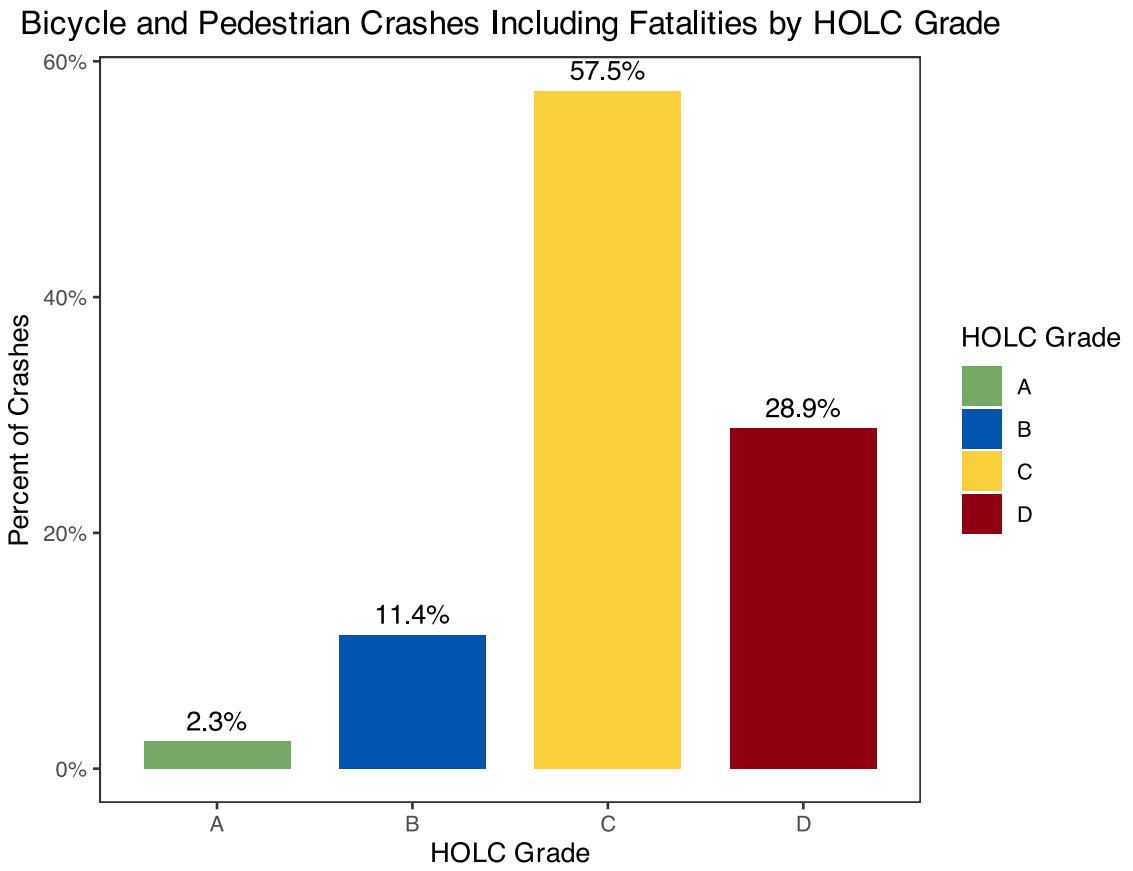
```
  mapping = aes(
    x = mean_pm25,
    y = mean_lle,
    color = grade
  )
) +
  geom_point(size = 3) +
  xlim(60, 100) +
  ylim(20, 60) +
  scale_color_manual(
    values = holc_colors
  ) +
  labs(
    title = "HOLC Particulate Matter 2.5 and Low Life Expectancy",
    x = "Average Particulate Matter 2.5 Percentile",
    y = "Average Low Life Expectancy Percentile",
    color = "HOLC Grade"
  ) +
  theme_bw() +
  theme(
    plot.title = element_text(hjust = 0.5)
  )
```



The scatterplot indicates that there is a positive relationship between average particulate matter 2.5 percentile and average low life expectancy percentile across HOLC grades. As historical HOLC grades decreased, average particulate matter 2.5 and low life expectancy percentiles increased. Census block groups assigned Grade D had the highest average percentiles for both indicators, while those assigned Grade A had the lowest percentiles.

```
# Summarizing bicycle and pedestrian crashes and fatalities by HOLC grade
bp_holc |>
  st_drop_geometry() |>
  group_by(grade) |>
  summarise(ped_bike_crash = n()) |>
  mutate(
    pct = ped_bike_crash / sum(ped_bike_crash) * 100
  ) |>
```

```
ggplot(
  # Creating bar graphs for all accidents (including fatalities)
  aes(
    x = grade,
    y = pct,
    fill = grade
  )
) +
geom_col(width = 0.75) +
geom_text(
  aes(label = paste0(round(pct, 1), "%")),
  vjust = -0.5
) +
scale_fill_manual(
  values = holc_colors
) +
scale_y_continuous(
  labels = scales::label_percent(scale = 1)
) +
labs(
  title = "Bicycle and Pedestrian Crashes Including Fatalities by HOLC Grade",
  x = "HOLC Grade",
  y = "Percent of Crashes",
  fill = "HOLC Grade"
) +
theme_bw() +
theme(
  plot.title = element_text(hjust = 0.5),
  panel.grid = element_blank()
)
```



Grade C had the highest percentage of bicycle and pedestrian crashes at 57.5%. The percentage of crashes in Grade D was 28.9%. Grades B and A had the lowest percentages at 11.4% and 2.3% respectively. Together, grades C and D accounted for 86.4% of bicycle and pedestrian crashes in graded neighborhoods, while containing about 77% of graded census block groups.

Conclusion

This analysis examined the neighborhoods in Los Angeles, California that were historically assigned HOLC grades to compare differences in socioeconomic and environmental indicators within Census block groups. Lower-graded neighborhoods had higher average low-income percentages, PM2.5 percentiles, and low life expectancy percentiles. For all three indicators, grade A neighborhoods had the lowest averages. Bicycle and pedestrian crashes were also disproportionately concentrated in lower-graded neighborhoods. These results indicate that there is an association between historical HOLC classifications and socioeconomic and environmental inequality. Redlining was a form of systemic oppression which has had long-lasting impacts on the health and economic stability of people of color [4]. Formerly redlined neighborhoods in Los Angeles experience disproportionate economic, environmental, and traffic safety injustices.

There are limitations to this analysis, due to many census block groups not having an HOLC grade. Results cannot be generalized across all of Los Angeles. Further analysis could examine

sidewalk access, bike lane availability, and street lighting to identify other factors that contribute to crashes. Results of further analysis may help inform future infrastructure planning.

References

[1] Nelson, R. K., Winling, L., et al. (2023). *Mapping inequality: Redlining in New Deal America*. Digital Scholarship Lab, University of Richmond. https://dsl.richmond.edu/panorama/redlining/howandwhy

[2] Mitchell, B., & Franco, J. (2018, March 20). *HOLC "redlining" maps: The persistent structure of segregation and economic inequality*. National Community Reinvestment Coalition. https://ncrc.org/holc/

[3] Schmidt, K. (2023, October 19). *Structural racism and pedestrian safety*. Public Health Post, Boston University School of Public Health. https://publichealthpost.org/health-equity/structural-racism-and-pedestrian-safety/

[4] Braveman, P. A., Arkin, E., Proctor, D., Kauh, T., & Holm, N. (2022). Systemic and structural racism: Definitions, examples, health damages, and approaches to dismantling. *Health Affairs, 41*(2), 171–178. https://doi.org/10.1377/hlthaff.2021.01394